Data Requirements
Drug Inventory Checker

This identifies the data required for the drug tracking system, the data types and sources and any given insights. For It all to be on one place.

- ⇨ Users are specified to be either patient or client, pharmacy or admin. User Accounts will have account data like login info, role, created at, last login, and contacts.

1. Patient

Data	Description	Type	Source	Notes
Personal info	Name, Date of birth, Location, Address	Strings, Text for Address, Integer for age, Location as GeoJSON Point	Input by user	Use a pin for location?
Medical info	Names of drugs used normally, frequent visited pharmacies	DrugID, PharmacyID	Input by user	Suggestion: the user will be prompted to list any drugs he take frequently upon register, and the names of pharmacies he normally purchase from (selected from a dropdown list)
Search History	Linked with searches id/Search terms	Array of Strings OR Array of SearchIDs	Handled at Backend	
Favorites	Linked with Favorite list of ids	Array of DrugIDs	Handled at Backend	

2. Pharmacies

Data	Description	Type	Source	Notes
Name	Name of the pharmacy	String	Input by pharmacy	
Logo	The URL for the logo	String	Input by pharmacy	Won't be stored as photo object to keep the DB lightweight
Location	The coordinates for the pharmacy	GeoJSON	API from the Google Maps	Will be used to calculate the nearest pharmacy location
Doctor in charge	The Administrator Doctor	String	API from the pharmacy page / input by user	Don't know if it exists on the db or not
Average rating		Float	Calculated at backend	Calculated aggregate from Rating table.
Inventory	List of drugs in stock	List of DrugIDs	API from the pharmacy DB	
Statues	Is it Active?	Boolean	Enabled or Disabled by admin	Inactive if not logged in for 30 days

Last Updated	The last update request	timestamp		To Ensure data is refreshed daily

3. Drugs

Data	Description	Type	Source	Notes
Generic Name	Name Representing active ingredients	String	API from Pharmacy DB	The actual name of the drug
Brand Name	The company labeled name for the drug which is marketed under	String	API from Pharmacy DB	
Pharmaceutical Company	The name of the Manufactural company	String	API from Pharmacy DB	
Category	Antibiotic, Analgesic, Antihistamine…etc	String	API from Pharmacy DB	
Price	Pharmaceutical Company set price	Float	API from Pharmacy DB	Prices for drugs are preset and not changed based on pharmacies
Patient Information Leaflet (PIL)	The Description of the drug. The one with the package	Text	API from Pharmacy DB	
Alternatives	Drugs with similar effect	Array of DrugIDs	Manual admin input	Who will input this?

Drugs and Pharmacies will be linked on the DB with an inventory table that has availability and amount available specifications

4. Searches

Data	Description	Type	Source	Notes
Patient	The patient who made this searchl	UUID (Universally Unique Identifier) for patient	Handled at backend	Need to be allowed null (if the patient who searched wasn't logged in)
Search Term	Text entered by the user	Text	Input by user as free text	Analysis will be based on the search terms
Search Type	How the user searched	Enum (Pharmacy, Category, Drug)	Input by user as radio buttons selection	
Time	When was the search made	Timestamp	Handled at backend	
Search Location	Where the search occurred (the location of patient)	GeoJSON Point	Linked with the Patient table	However the patient location was collected (Free text or a bin with API from Google Maps)
# Results Returned	Number of results returned from that search	Integer	Handled at backend (all valid returns)	

5. Notifications

We have multiple types of notifications triggered by multiple events

Data	Description	Type	Source	Notes
User	The User to receive the notification	UUID	Linked with Patient/Pharmacy table	
Type	Drug Available, Price Change, Frequent Demand, Unavailable	Enum of strings	Determined by event	
Message	Notification content	Text	Hardcoded	
Statues	Is it read?	Boolean	Handled at Backend	
Created at	When it was triggered.	Timestamp		

6. Favorites

Favorites will be a simple list created in backend with specifying each the drug, the user, and the statues of notification sent to ensure not repeating notifications

Favorites must be linked with notifications for any updates on the drugs on the list.

7. Ratings

Handles patient feedback on pharmacies, Also handled with a simple table in the backend with patient, pharmacy, value, comments, and timestamp.

8. Analytical Data

Data that serves analytical purposes (we have multiple) should be precomputed so performance won't drastically decrease, In an Architecture and design view all the following data won't necessarily on the same physical table or storage system. For Drugs:

Data	Description	Type	Source	Notes
Drug	The specific drug for the report	DrugID	API from DB	
Average Price	The average price over all pharmacies	Float	Calculated Aggregate from DB	
Availability	Percentage of pharmacies that have it in stock	Float	Calculated Aggregate from DB	
Search Count	Times it was returned as a search result	Integer	Calculated	
Date of computation	Date it was calculated	Date		Important to ensure reports are up to date

And For pharmacies:

Data	Description	Type	Source	Notes
Pharmacy	The specific pharmacy for the report	PharmacyID	API from DB	Pharmacies get to view those analytics
Search Count	Number of searches with the pharmacy name	Integer	Calculated using "Search Term" in Searches table	
Page Views	Number of people that clicked the pharmacy page	Integer	Calculated from raw event logs	Clicks are stored as events and calculated daily for this field

Date of computation	Date it was calculated	Date		Important to ensure reports are up to date

Pharmacies get some link with drugs searches that don’t return results and the number of searches.

Note: this is not a replacement for ERD in any way.